Industrial Internship Report on

”Quality Prediction in a Mining Process”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 4 weeks' time. My project was to **predict the percentage of silica concentrate (% Silica Concentrate)** in a mining flotation process using **Machine Learning**. Predicting silica levels in advance helps engineers improve ore quality, reduce impurities, minimize iron loss, and optimize plant operations. The project uses a real-world mining dataset containing process parameters such as **Iron Feed, Silica Feed, Ore Pulp Flow, pH, Density, Air Flow, and Flotation Column Levels**. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

This internship project aims to predict % Silica Concentrate in the Iron Ore Flotation Process using Machine Learning techniques. Prediction of % Silica Concentrate which is an undesired material in the mining process helps in optimizing the production process and reducing loss of material.

This project was conducted over a period of 4 weeks on a real-world data set provided by UniConverge Technologies Pvt. Ltd. This project involved several major tasks like EDA (Exploratory Data Analysis), Preprocessing, Feature Engineering, Model Building & Training, Hyperparameter Tuning, and finally, Evaluation of results. Linear Regression, Decision Tree Regressor, and Random Forest Regressor were used to build predictive models. The best-fitted model was then used for prediction of % Silica Concentrate without using % Iron Concentrate. Apart from this, the project also involved exploring the prediction of % Silica Concentrate after one hour to enable optimization of the mining process.

In conclusion, this internship experience helped me gain a lot of practical insight into the field of Machine Learning, in particular, how the concepts are applied in solving real-world problems, like industrial manufacturing, which in turn, helps in better comprehension of the subject.

Industrial internships help bridge the gap between classroom education and practical implementation, thereby exposing students to real-world scenarios and enhancing their knowledge and skills.

During my internship at upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd, I was exposed to various aspects of industrial projects. The project that I undertook viz. "Quality Prediction in a Mining Process" aimed at developing a Machine Learning model that could predict % Silica in Iron Ore. This involved using historical and operational data from a mining flotation plant to train and evaluate multiple machine learning models. The insights obtained from the project were useful in improving my knowledge and skills in Data Science and Machine Learning.

I extend my sincere gratitude to upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd, my faculties and well-wishers for their support, guidance and encouragement throughout the internship program.

I would like to acknowledge the support and guidance rendered by upskill Campus, The IoT Academy and UniConverge Technologies Pvt. Ltd, throughout the internship program.

I also express my sincere gratitude towards my college ITER, Siksha 'O' Anusandhan (SOA) and faculties for their constant support and guidance throughout the internship.

Finally, I thank my family members and well-wishers for motivating me to complete the internship and successfully accomplish the project.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

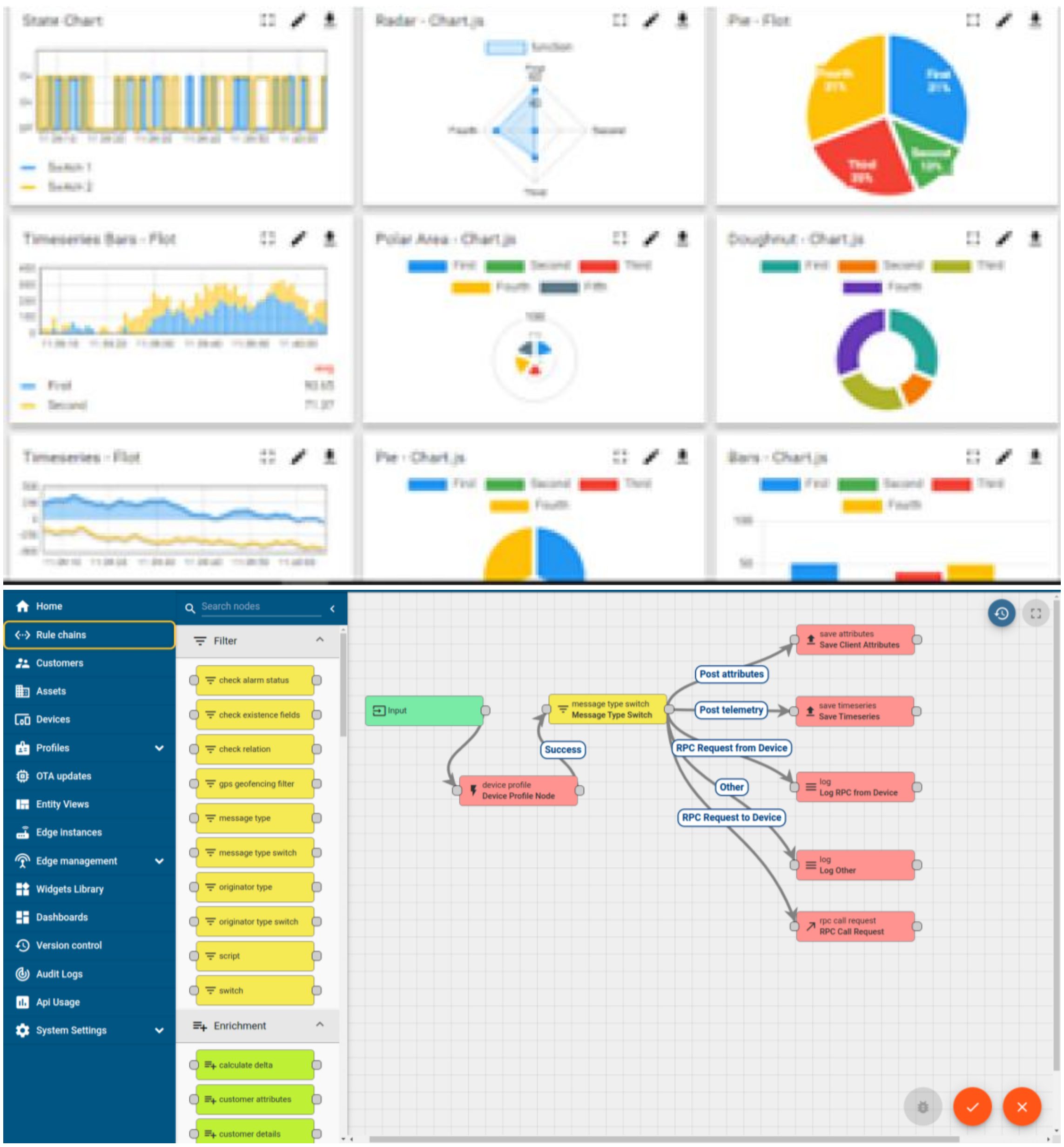
i. UCT IoT Platform ( )

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



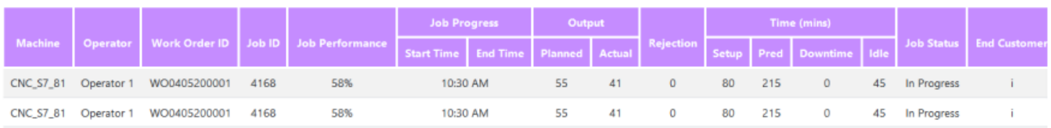
ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



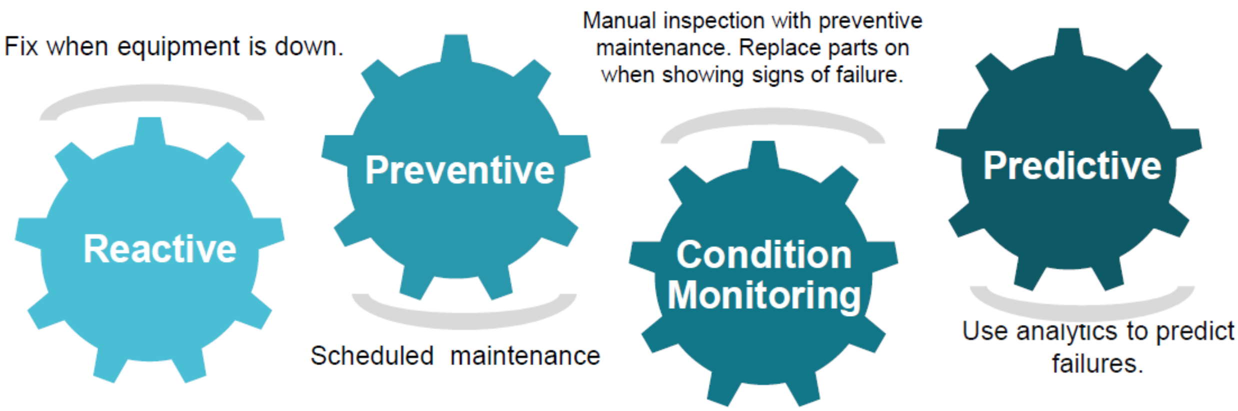
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

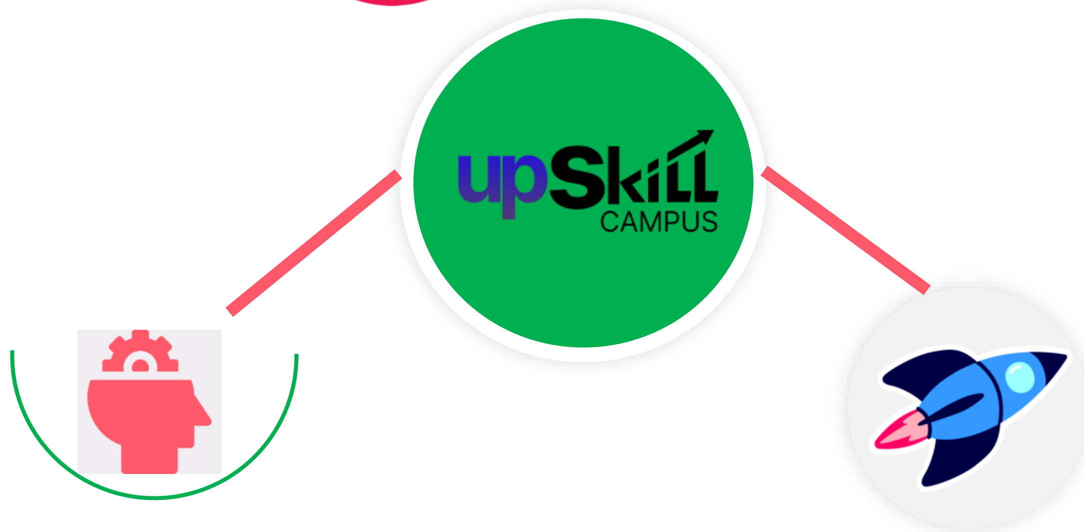
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

- Career growth/upskilling
  - Interview Preparation and skill building
  - upskilling Courses
  - Skill Assessment
  - Profile building
- Professional networking
  - Alumni Connections
  - Mentorship
  - Discussion/QA forum
- Collaboration platform
  - Project collaboration
  - Discussion forum
  - Tech updates
- Job/internship platform
  - Job portal
  - Internship portal
  - Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] UCI Machine Learning Repository, **"Machine Learning Repository."** Available: https://archive.ics.uci.edu/

[2] Pedregosa, F., Varoquaux, G., Gramfort, A., et al., **"Scikit-learn: Machine Learning in Python,"** *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[3] Wes McKinney, **Python for Data Analysis**, 3rd Edition, O'Reilly Media, 2022.

[4] NumPy Developers, **"NumPy Documentation."** Available: https://numpy.org/doc/

[5] Pandas Development Team, **"Pandas Documentation."** Available: https://pandas.pydata.org/docs/

[6] Matplotlib Development Team, **"Matplotlib Documentation."** Available: https://matplotlib.org/stable/

[7] Seaborn Development Team, **"Seaborn Documentation."** Available: https://seaborn.pydata.org/

[8] Breiman, L., **"Random Forests,"** *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[9] Géron, A., **Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow**, 3rd Edition, O'Reilly Media, 2022.

2.6 Glossary

Term	Abbreviation
Artificial Intelligence	AI
Machine Learning	ML
Exploratory Data Analysis	EDA
Random Forest	RF
Linear Regression	LR
Decision Tree	DT
Mean Absolute Error	MAE
Root Mean Square Error	RMSE
Coefficient of Determination	R^2
Hyperparameter Tuning	HPT
Flotation Process	FP
Iron Ore Concentrate	IOC
Silica Concentrate	SC
Feature Engineering	FE

3 Problem Statement

In the assigned problem statement

The mining industry relies on the flotation process to separate valuable iron ore from unwanted impurities. One of the key quality indicators of the final product is the **percentage of silica concentrate (% Silica Concentrate)**. A lower silica concentration indicates higher-quality iron ore and improved production efficiency.

In conventional mining operations, the silica concentration is measured through laboratory analysis after the flotation process is completed. Since laboratory testing requires time, engineers are unable to make immediate adjustments to the process, which may lead to increased impurities, reduced iron recovery, and higher material losses.

The objective of this project is to develop a **Machine Learning-based predictive model** that estimates the **% Silica Concentrate** using real-time process parameters collected from a flotation plant. The dataset contains operational variables such as **Iron Feed, Silica Feed, Starch Flow, Amina Flow, Ore Pulp Flow, Ore Pulp pH, Ore Pulp Density, Air Flow, and Flotation Column Levels**.

The developed model enables engineers to predict silica concentration before laboratory results become available. This helps in taking proactive corrective actions, improving ore quality, reducing impurities, minimizing environmental waste, and optimizing overall plant performance.

The project also evaluates multiple machine learning algorithms, including **Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**, to identify the most accurate prediction model. Additionally, experiments were conducted to predict silica concentration without using the **% Iron Concentrate** feature and to perform **one-hour-ahead prediction** for predictive process optimization.

4 Existing and Proposed solution

Existing Solution:

In conventional mining flotation plants, the quality of iron ore concentrate is monitored by measuring the percentage of silica concentrate through laboratory analysis. Since laboratory testing is performed after the flotation process, engineers receive the results with a significant time delay. This delay prevents immediate corrective actions, leading to reduced process efficiency, increased impurities, higher iron losses, and unnecessary wastage of valuable resources.

Traditional monitoring methods mainly depend on manual observation and historical process data, which are often insufficient for making real-time decisions. As a result, maintaining consistent product quality becomes challenging, especially under changing operating conditions.

Limitations of Existing Solution

- Laboratory testing is time-consuming and cannot provide real-time feedback.
- Delayed results reduce the ability to make immediate process adjustments.
- Manual monitoring depends heavily on operator experience.
- High silica concentration may lead to poor iron ore quality.
- Increased operational costs due to inefficient process control.
- No predictive capability for future silica concentration.

Proposed Solution:

To overcome these limitations, a **Machine Learning-based prediction system** has been developed to estimate the **% Silica Concentrate** using real-time operational parameters collected from the mining flotation plant.

The proposed system performs data preprocessing, feature engineering, and model training using historical process data. Multiple regression algorithms, including **Linear Regression**, **Decision Tree Regressor**, and **Random Forest Regressor**, were implemented and evaluated. Among these, the **Random Forest Regressor** achieved the highest prediction accuracy and was selected as the final model.

The proposed solution enables engineers to predict silica concentration before laboratory measurements become available, allowing timely corrective actions that improve production efficiency and reduce material losses.

Value Addition of the Proposed Solution

- Provides accurate prediction of **% Silica Concentrate** using Machine Learning.
- Supports proactive decision-making before laboratory results are available.
- Reduces impurities and improves iron ore quality.
- Minimizes iron losses and environmental waste.
- Improves overall operational efficiency of the flotation process.
- Demonstrates that silica concentration can be predicted even without using the **% Iron Concentrate** feature.
- Supports future prediction to assist engineers in process optimization.

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

The proposed system utilizes the Machine Learning pipeline for silica concentrate percentage prediction in the mining company. Initially, the data was gathered from the industrial source and preprocessed to represent the required format for the ML algorithm usage. A set of exploratory data analysis (EDA) was conducted to understand the structure and feature connection. Additionally, the data was analyzed for possible errors or abnormalities.

The feature engineering stage was applied to the finalized data set, and the model was trained and tested on the selected samples. Several regression algorithms were trained, and the random forest regressor showed the best performance among the others and was chosen for the prediction task in the mining company. The trained model is able to output the approximate percentage of the silica concentrate, which helps the engineers to take the necessary action before the lab results are available.

5.1 High Level Diagram

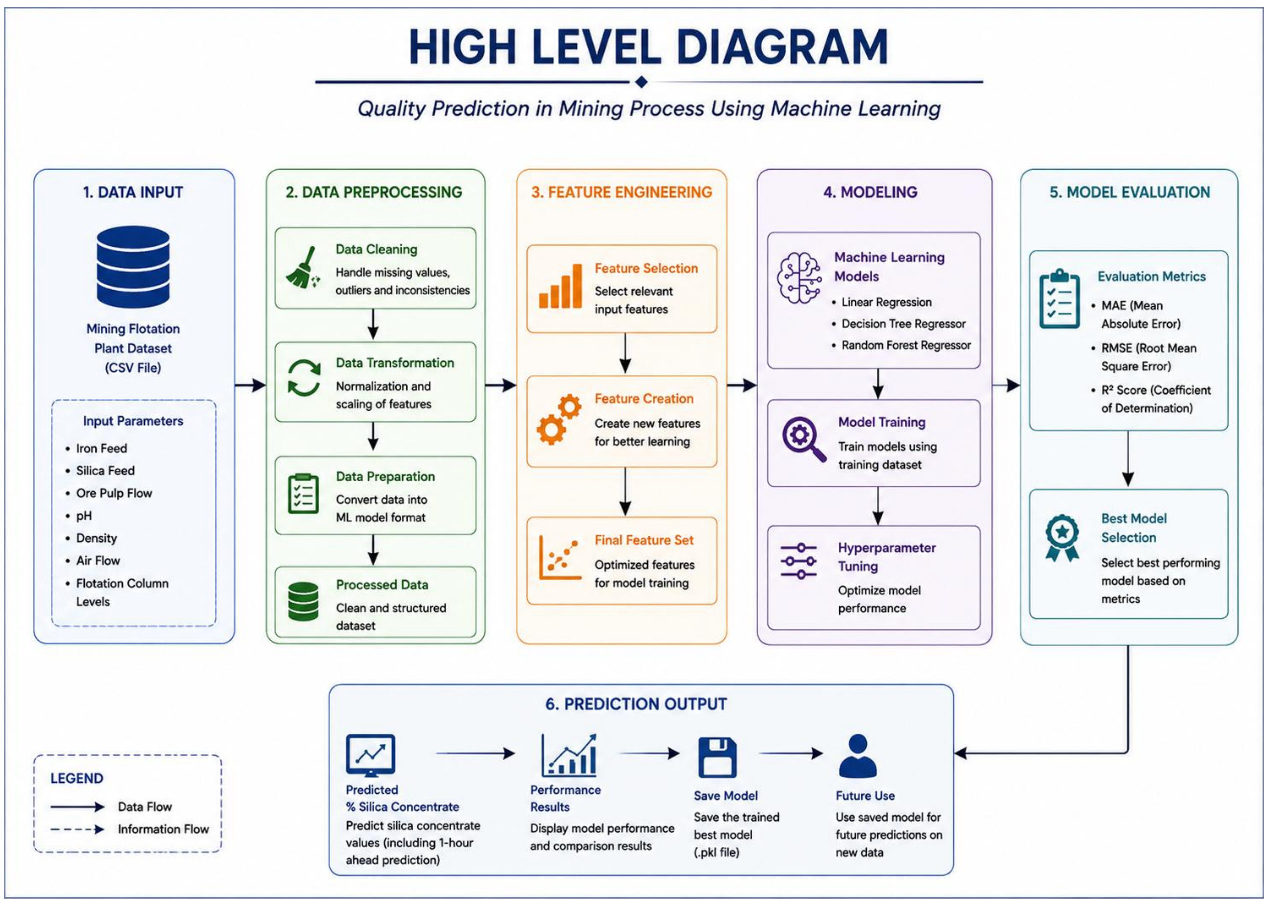


Fig 5.1.1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

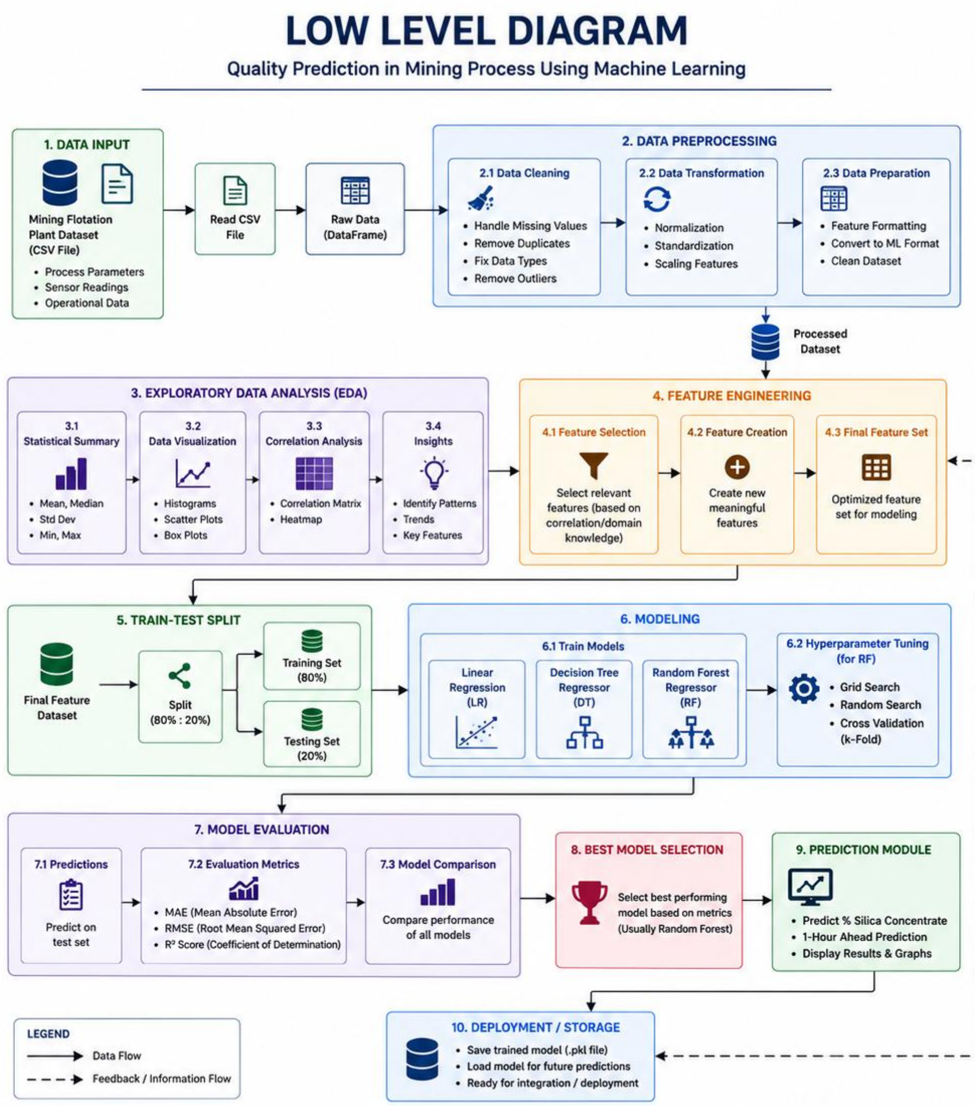


Fig 5.1.2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

The proposed system consists of the following logical modules:

Input Module

- Industrial Mining Dataset
- Process Parameters
- Sensor Data

Processing Module

- Data Cleaning
- Feature Engineering
- Machine Learning Model Training
- Hyperparameter Optimization

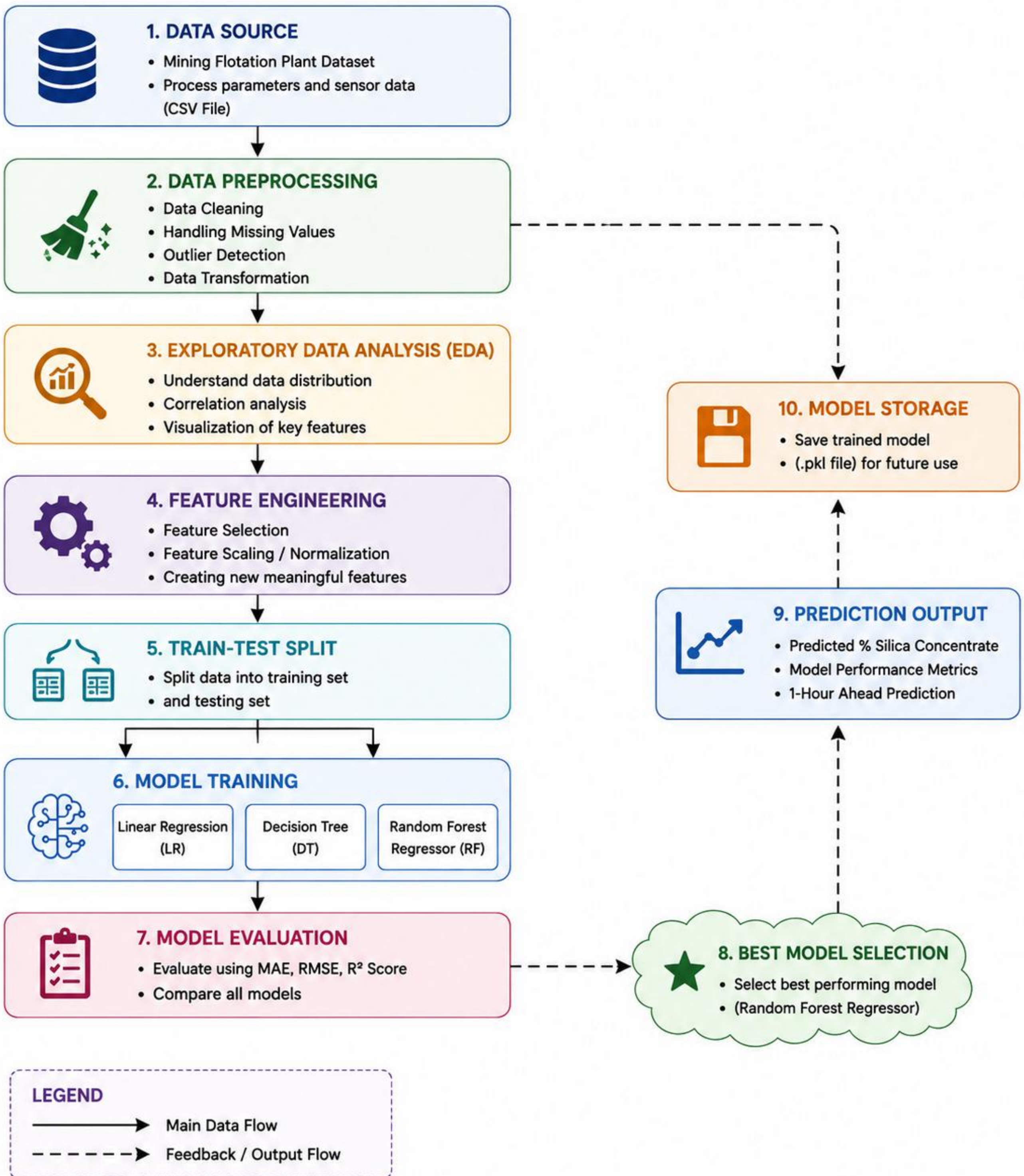
Output Module

- Predicted **% Silica Concentrate**
- Model Performance Metrics (MAE, RMSE, R^2 Score)
- Feature Importance Analysis
- Future Prediction Results

System Workflow

1. Collect industrial mining process data.
2. Preprocess and clean the dataset.
3. Perform Exploratory Data Analysis (EDA).
4. Engineer relevant features for model training.
5. Train multiple regression models.
6. Evaluate model performance using MAE, RMSE, and R^2 Score.
7. Select the best-performing model (Random Forest).
8. Predict silica concentration for new process data.
9. Save the trained model for future deployment.

DATA FLOW DIAGRAM



6 Performance Test

The developed Machine Learning system was evaluated to determine its effectiveness in predicting the **% Silica Concentrate** in a mining flotation process. The evaluation focused on identifying practical constraints that affect industrial deployment and assessing how the proposed solution addresses them. The major constraints considered during the project include:

- **Prediction Accuracy:** The model should provide highly accurate predictions to support reliable decision-making.
- **Processing Speed:** The prediction process should be fast enough to support near real-time monitoring.
- **Large Dataset Handling:** The system should efficiently process over **736,000 records** without significant performance degradation.
- **Model Generalization:** The model should avoid overfitting and maintain good performance on unseen data.
- **Scalability:** The solution should be capable of handling larger datasets and future deployment in industrial environments.

These constraints were addressed by performing proper data preprocessing, feature engineering, train-test splitting, hyperparameter tuning, and selecting the best-performing regression model.

6.1 Test Plan/ Test Cases

Test Case ID	Test Description	Expected Result	Actual Result	Status
TC-01	Load Mining Dataset	Dataset loads successfully	Successfully Loaded	Pass
TC-02	Data Preprocessing	Data cleaned without errors	Successfully Completed	Pass
TC-03	Exploratory Data Analysis	Statistical insights generated	Successfully Completed	Pass
TC-04	Train Regression Models	Models trained successfully	Successfully Completed	Pass
TC-05	Hyperparameter Tuning	Improved model performance	Successfully Completed	Pass
TC-06	Predict % Silica Concentrate	Accurate prediction generated	Successfully Completed	Pass
TC-07	Future Prediction	Predict one-hour-ahead silica concentration	Successfully Completed	Pass
TC-08	Save Final Model	Model saved successfully (.pkl)	Successfully Completed	Pass

6.2 Test Procedure

The performance evaluation was carried out using the following procedure:

1. The mining flotation dataset was loaded into the Python environment.
2. Data preprocessing was performed to clean and transform the dataset.
3. Exploratory Data Analysis (EDA) was conducted to understand data distribution and feature relationships.
4. Feature engineering techniques were applied to improve model performance.
5. The dataset was divided into training and testing sets using an 80:20 split.
6. Three regression models—Linear Regression, Decision Tree Regressor, and Random Forest Regressor—were trained.
7. Hyperparameter tuning was applied to optimize the Random Forest model.
8. The models were evaluated using **Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score**.
9. The best-performing model was selected and used for predicting silica concentration.
10. The final model was saved for future deployment.

6.3 Performance Outcome

The performance of the proposed machine learning system was evaluated using standard regression metrics, including **Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score**. Multiple regression models, namely **Linear Regression, Decision Tree Regressor, and Random Forest Regressor**, were trained and compared to identify the most suitable model for predicting the **% Silica Concentrate**.

Among all the models, the **Random Forest Regressor** demonstrated the best overall performance, achieving the highest prediction accuracy and the lowest prediction error. Hyperparameter tuning was further applied to optimize the model and improve its generalization capability. An additional experiment was conducted by removing the **% Iron Concentrate** feature, and the model still maintained excellent predictive performance, proving that the remaining process parameters contain sufficient information for accurate prediction. The project also successfully demonstrated the feasibility of predicting silica concentration **one hour in advance**, enabling proactive process monitoring and decision-making.

The screenshots below present the performance metrics, model comparison results, feature importance analysis, and prediction outputs obtained during the implementation. These results confirm that the proposed machine learning solution is accurate, reliable, and suitable for supporting quality prediction in real-world mining flotation processes.

	Model	MAE	RMSE	R2 Score
0	Linear Regression	0.482440	0.626604	0.689001
1	Decision Tree	0.001997	0.038824	0.998806
2	Random Forest	0.003017	0.020358	0.999672

Fig 6.3.1 Model comparison

```
Optimized Random Forest
----------------------------
MAE : 0.017770912561595787
RMSE: 0.03575002153657188
R² : 0.9989876631759009
```

Fig 6.3.2 Optimized Random Forest result

```
Random Forest WITHOUT Iron Concentrate
-------------------------------------
MAE : 0.013030476122038183
RMSE: 0.06302173741127017
R² : 0.996854040242823
```

Fig 6.3.3 Random Forest result without Iron concentrate

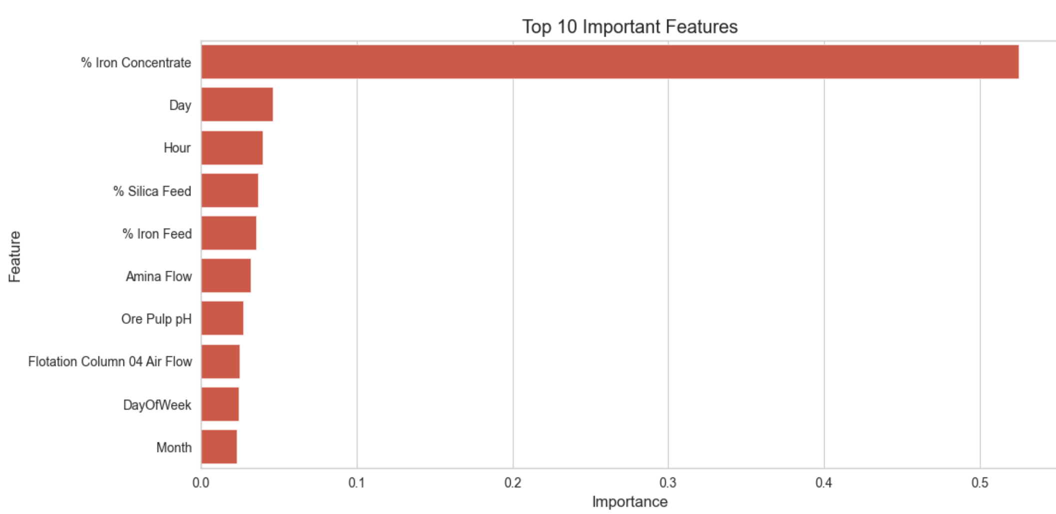


Fig 6.3.4 Top 10 features

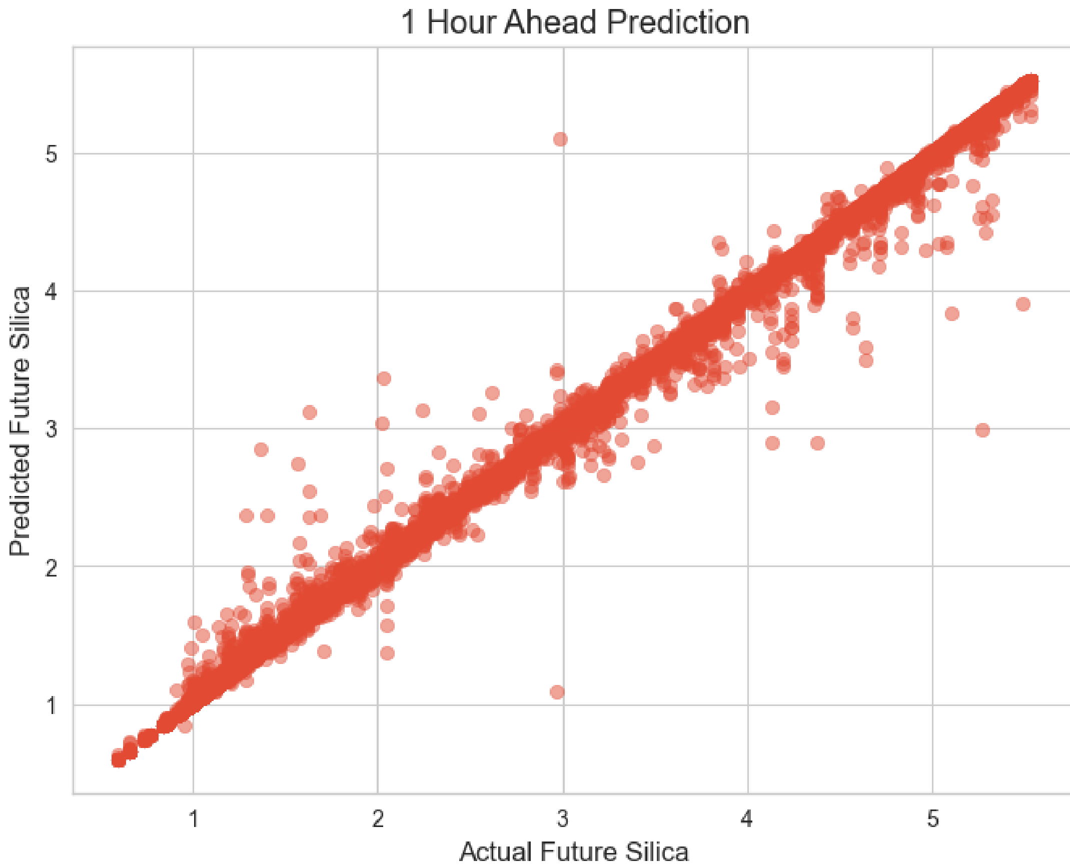


Fig 6.3.5 ACTUAL VS FUTURE PREDICTION

	Model	MAE	RMSE	R^2 Score
0	Linear Regression	0.482440	0.626604	0.689001
1	Decision Tree	0.001997	0.038824	0.998806
2	Random Forest	0.003017	0.020358	0.999672
3	Optimized Random Forest	0.017771	0.035750	0.998988
4	Random Forest (Without Iron)	0.013030	0.063022	0.996854
5	Future Prediction Model	0.003295	0.025386	0.999490

Fig 6.3.6 Final Results

7 My learnings

This internship gave me an insight into the practical application of Machine Learning in the industry. During this project, I got familiar with working on a large industrial data set, performing data preprocessing, data exploration, data visualization, data engineering, and predictive modeling in Python.

I enhanced my Python coding skills while working on Pandas, NumPy, Matplotlib, and Seaborn libraries for data science tasks such as data preprocessing and data visualization. I also learned how to leverage Scikit-learn's machine learning algorithms for model training and hyperparameter tuning, evaluate model performance using evaluation metrics such as MAE, RMSE, and R2 Score, and apply predictive analytics in the industry to make data-driven decisions.

This internship experience also helped me to develop my problem-solving, analytical, and programming skills. The skills that I gained from this project will propel my future career in Machine Learning, Artificial Intelligence, Data Science, and Industrial Analytics.

8 Future work scope

Although the proposed Machine Learning model demonstrated outstanding accuracy, there are still some ideas that could be further explored and implemented. The four-week project timeline did not allow the team to consider all details and possibilities for improvement. Some of the most interesting ideas for future development and research include

upgrading the model to process the sensor data in real-time and utilizing more advanced Machine Learning and Deep Learning algorithms such as XGBoost, LightGBM, LSTM, GRU, etc.

The model can also be presented as a web dashboard or cloud application that will be used to view and analyze the prediction results. Moreover, the proposed solution can be expanded to include other elements of the mining plant, such as IIoT (Industrial Internet of Things) and SCADA (Supervisory Control and Data Acquisition) systems, which will allow it to operate in real-time as well. Predictive maintenance and failure detection systems can also be implemented in the future to ensure the continuous and uninterrupted operation of the mining plant and, thus, make it more cost-effective.